

HOUSTON HOBBY, TX, USA

WMO: 722440

Lat: 29.638N

Lon: 95.282W

Elev: 13

StdP: 101.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	2.6	-8.6	1.8	6.4	-6.2	2.2	8.3	11.2	10.9	10.2	11.9	4.2	0	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	35.7	25.1	34.7	25.2	33.8	25.1	26.9	31.5	26.6	31.2	26.3	30.8	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	21.4	28.4	25.5	20.8	28.2	25.1	20.3	28.1	84.5	31.4	82.7	31.0	81.4	30.8	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.6	-2.1	37.7	2.0	1.6	-3.5	38.9	-4.7	39.9	-5.8	40.8	-7.2	42.0		
			-4.0	27.9	1.7	1.0	-5.2	28.6	-6.2	29.2	-7.2	29.7	-8.5	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.7	12.8	14.8	17.9	21.3	25.3	28.1	29.1	29.3	27.1	22.7	17.5	13.7
	DBStd	7.00	5.22	5.16	4.36	3.31	2.51	1.66	1.39	1.58	2.36	3.79	4.73	5.18
	HDD10.0	77	32	16	4	0	0	0	0	0	0	0	4	21
	HDD18.3	619	184	118	62	12	1	0	0	0	0	10	71	161
	CDD10.0	4334	117	151	248	339	473	544	592	599	514	392	229	137
	CDD18.3	1836	11	20	48	102	216	294	334	341	264	144	46	18
	CDH23.3	17979	27	63	184	579	1856	3243	3929	4015	2652	1127	240	65
	CDH26.7	7188	0	4	17	98	604	1392	1792	1872	1049	324	32	3
Wind	WSAvg	3.5	4.0	4.1	4.1	4.1	3.7	3.1	2.7	2.6	2.8	3.2	3.5	3.8
Precipitation	PrecAvg	1340	100	84	74	84	130	178	113	113	137	118	110	97
	PrecMax	2109	243	197	201	266	377	589	438	407	381	318	259	228
	PrecMin	677	5	11	10	2	0	4	2	17	14	0	10	12
	PrecStd	380	58	52	47	68	88	131	86	84	80	89	72	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	27.2	28.5	30.5	33.7	36.4	36.7	37.4	35.8	32.7	29.3	27.1
		MCWB	19.2	19.6	20.2	21.6	23.7	24.4	25.1	25.1	24.3	24.1	22.8	21.1
	2%	DB	24.0	25.2	26.8	28.9	32.2	34.4	35.2	35.9	34.0	31.3	27.5	25.1
		MCWB	19.0	19.5	20.3	21.6	23.8	24.8	25.4	25.4	24.5	23.8	21.6	20.9
	5%	DB	22.6	23.7	25.5	27.7	31.1	33.3	34.2	34.7	32.8	29.9	26.1	23.6
		MCWB	18.8	19.2	19.9	21.1	23.7	24.8	25.4	25.4	24.4	23.1	21.1	20.3
	10%	DB	21.1	22.2	24.0	26.6	30.0	32.4	33.4	33.7	31.8	28.5	24.5	22.1
		MCWB	18.2	18.8	19.3	20.7	23.5	24.8	25.3	25.4	24.2	22.2	20.6	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	22.2	22.9	24.5	26.5	27.3	27.5	27.5	27.0	26.2	24.3	23.0
		MCDB	23.3	24.1	25.7	27.9	30.9	31.7	32.3	32.3	30.8	29.1	26.9	24.9
	2%	WB	20.8	21.3	22.0	23.6	25.6	26.7	26.9	26.9	26.4	25.5	23.4	22.0
		MCDB	22.5	23.4	24.5	26.5	29.5	30.9	31.7	31.8	30.1	28.7	25.9	23.9
	5%	WB	20.0	20.6	21.4	22.9	25.0	26.2	26.4	26.6	25.9	24.9	22.6	21.2
		MCDB	21.8	22.6	23.8	25.7	28.7	30.3	31.4	31.5	29.5	28.1	25.0	23.1
	10%	WB	18.8	19.8	20.8	22.4	24.5	25.7	26.1	26.2	25.5	24.1	21.5	19.9
		MCDB	20.7	21.7	23.0	25.1	28.1	29.9	30.8	31.0	29.2	27.2	23.8	21.8
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.2	9.3	9.2	8.7	8.4	8.6	8.8	8.7	9.5	9.8	9.4
		MCDBR	10.8	9.8	9.4	9.4	9.5	9.6	9.7	10.2	9.6	9.5	9.8	10.2
		MCWBR	7.1	5.9	4.8	4.3	3.4	2.7	2.7	2.6	3.0	3.8	5.3	6.7
	5% WB	MCDBR	9.0	8.3	7.9	7.5	8.1	8.0	8.7	8.9	7.9	8.1	8.7	9.2
		MCWBR	7.1	6.1	5.2	4.3	3.5	3.0	2.9	2.8	3.0	4.4	5.9	7.1
Clear-Sky Solar Irradiance	taub	0.343	0.361	0.375	0.439	0.482	0.472	0.500	0.487	0.458	0.381	0.372	0.352	
	taud	2.504	2.463	2.414	2.232	2.150	2.228	2.172	2.209	2.258	2.465	2.438	2.485	
	Ebn at Noon	893	905	912	859	821	823	800	808	820	871	850	862	
	Edh at Noon	90	102	114	142	154	142	150	143	132	101	96	88	
All-Sky Solar Radiation	RadAvg	2.88	3.25	4.45	5.31	5.93	6.15	5.97	5.73	4.97	4.34	3.28	2.56	
	RadStd	0.28	0.56	0.38	0.38	0.42	0.49	0.56	0.47	0.52	0.52	0.33	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	+112
(47)	Regional (35 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+97

Nomenclature: See separate page